

Nakul Khandelwal

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ML Engineer — computer vision, inference systems, end-to-end pipelines

EXPERIENCE

Data Scientist

July 2024 – Present

Stylumia

Bengaluru, India

- Solely owned Stylumia's **Trends Engine** end-to-end spanning clustering, scoring, forecast, and naming, improving hit rate from ~2% to 78% through rapid client-driven iteration **across retail, social, and runway** verticals
- Built a **runway intelligence pipeline** for **H&M** with LLM-based garment attribution at **93% accuracy** on untagged pieces, aggregating color, fabric, and silhouette signals across **80–90 shows** within a **36-hour SLA**
- Designed **LLM-as-a-judge** evaluation infrastructure for subjective modules, reducing output failure rate from **78% to 54%** in three iterations; actively detects model-upgrade regressions and drives prompt adaptation
- Architected **Dockerized SQS** workflows and forward-compatible **Elasticsearch** schemas across trends and runway, **powering weekly client-facing deliveries** and enabling scalable adoption across enterprise clients globally
- Engaged directly with clients and cross-functional teams through **sales calls, feedback loops, PRD discussions, and UX collaborations**, translating client feedback into pipeline and scoring changes

ML Engineer

July 2023 – December 2023

Vehant Technologies

Noida, India

- Identified **CIM** (RGB-D saliency architecture) as an **ONNX**-convertible replacement for threat detection, confirmed accuracy parity, and validated export via **ONNXRuntime** and **OpenCV** for embedded deployment
- Built an **X-ray image annotation pipeline** using **SLIC** superpixel segmentation integrated with a custom **GUI**, enabling efficient, intuitive labeling and scalable training data preparation for threat detection model training
- Elevated **X-ray baggage scanner** performance to meet **ECAC standards**, securing **Delhi Airport** installation

Software Engineer

May 2022 – July 2022

Army Base Workshop

Remote

- Developed an **offline-capable** drone tracking application for the Indian army that relied on GPS and local data
- Implemented software features as a part of a team, to provide **real-time actionable intelligence** to military

PROJECTS

CV Edge Inference Benchmark | *Edge Inference, Computer Vision*

May – Sep 2026

- Benchmarked **YOLOv8n** on PyTorch, ONNX Runtime, TensorRT (FP32/FP16/INT8) across CPU, T4, and M5
- Live webcam demo ran **58% above benchmark latency**, traced via CPU clock counters to idle downclocking
- **INT8** ran only **6% faster** than FP16 on T4 at **18.4% lower mAP**, making FP16 the deployment choice

Feature-Augmented Visual Clustering | *Computer Vision & ML*

October 2024

- Applied **CLIP embeddings** on segmented retail garments to capture attributive similarity over raw visual texture
- Agglomerative clustering with **silhouette score optimisation** cut polluted groups from **63% to 17%**

Sudoku Reasoning Lab | *ML Systems, Constraint Satisfaction*

March 2026

- Typed **constraint propagation engine**, within a **CSP** environment with 5-tier difficulty classification, 234 tests
- External solver subsumed our propagation stack, exposing a divergence in constraint reasoning architecture

EDUCATION

Birla Institute of Technology And Science, Pilani (BITS Pilani)

Pilani, Rajasthan

Bachelor of Engineering (Hons.) in Electrical and Electronics | 6.85/10

September 2020 – June 2024

JEE Advanced: 7006 | **JEE Main:** 1663 | **BITSAT:** 344/400

TECHNICAL SKILLS

Inference & Deployment: PyTorch, ONNX Runtime, TensorRT, OpenCV

Infrastructure & Languages: Docker, SQS, Elasticsearch, PostgreSQL, Redis, Python, C++, TypeScript, SQL

Coursework: Modern Robotics, Control Systems, Neural Networks, Machine Learning, Data Structures and Algorithms

POSITIONS OF RESPONSIBILITY

Team Lead | *Dance Club*: Led 20 dancers, held competitions with 15+ colleges, and won 2nd place at IIM Kozhikode

Mentor | *Astro Club*: Presented and learnt the RA/DEC system, and contributed to Astro-42 quiz success